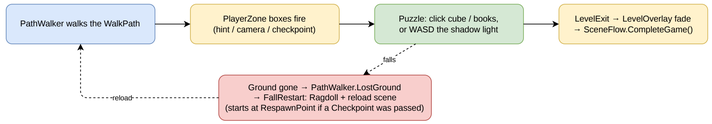
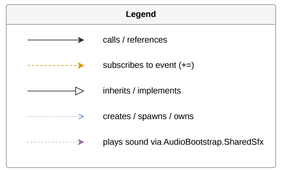
Smoking — Technical Design

Scene flow (GameSettingsConfig.levelScenes, SceneFlow.CompleteGame)



Inside a level





Player System Smoking.Player

PathWalker
MonoBehaviour · on "Player"
vars
WalkPath path
float speed = 1.4
EndMode endMode (PingPong / Loop / Stop / WalkOff)
float maxStepHeight, maxSlope, probeAhead
float retreatDistance
bool startWherePlaced (editor)
float distance // along path
int direction
bool stopped
float groundHeight
functions / events
event Action LostGround
Vector3 Velocity
Start() → start at checkpoint / path start
Update() → move along path, raycast ground
IsWallAhead() → turn around + retreat
TryGround(point, out y)
HandlePathEnd()

FallRestart
MonoBehaviour · on "Player"
vars
PathWalker walker
Ragdoll ragdoll
float restartDelay = 2.5
AudioCue fallSound
functions / events
HandleLostGround()
RestartAfterDelay() → reload active scene

Ragdoll
MonoBehaviour · on MannequinModel
vars
float totalMass
List<Rigidbody> bodies
List<Collider> colliders
bool IsActive
functions / events
Awake() → Build() joints from Humanoid bones
Activate(velocity)
TryGet() in Start
Ics(on)

WalkPath
MonoBehaviour · child Transforms = waypoints
vars
List<Vector3> points
float Length
functions / events
Rebuild()
Evaluate(distance, out forward)
ClosestDistance(point)

RespawnPoint
static class · survives scene reload
vars
static string scene
static Vector3 position
functions / events
Set(point)
Clear()
TryGet(out point)

Evaluate() / ClosestDistance()
+= LostGround
Activate(velocity)
Clear()

Game Flow / Scene Management Smoking.GameFlow

SceneFlow
MonoBehaviour · one per scene
vars
AudioCue backgroundMusic, clickSound
bool loading
functions / events
StartGame() / StartLevel(index)
RestartGame() / MainMenu() / Credits()
CompleteGame() → NextScene
Load(scene) → RespawnPoint.Clear(), LoadSceneAsync
Quit()

PlayerZone
abstract MonoBehaviour
vars
Vector3 areaSize
Transform player
bool inside
SharedSfx.Play
functions / events
Update() → polls player position in box
abstract OnPlayerEnter()
virtual OnPlayerExit()

LevelOverlay
Singleton · Resources/LevelOverlay prefab
vars
CanvasGroup hintGroup, fadeGroup, titleGroup, quoteGroup
TMP_Text hintText, title, subtitle, quoteText
float charactersPerSecond
functions / events
static Instance
ShowHint(owner, message)
HideHint(owner)
PlayTransition(heading, caption, quote, author, fade, hold)

GameSettingsService
Singleton · DontDestroyOnLoad
vars
static Instance
GameSettingsConfig config
AudioBootstrap audioRoot
float Brightness
functions / events
event Action<float> BrightnessChanged
SetBrightness(value)
Volume / Music / Sfx (from AudioBootstrap)
Save()

LevelExit
PlayerZone
vars
SceneFlow flow
string heading, caption
float fadeTime, holdTime
UnityEvent reached
functions / events
OnPlayerEnter() → stop walker
Finish() → PlayTransition, CompleteGame()

GameSceneBuilder
Editor · Tools/Smoking/Create Missing Game Scenes
functions / events
Build() — creates MainMenu / gameplay / Credits scenes
wires SceneFlow, GameSettingsService, MenuPanels, SettingsPanel, GameplayClock

GameSettingsConfig
ScriptableObject · GameSettings.asset
vars
string mainMenuScene
string[] levelScenes
string creditsScene
string[] levelQuotes, quoteAuthor
float minimumBrightness, defaultBrightness
functions / events
SceneAfter(current)
QuoteAfter(level)

Checkpoint
PlayerZone · Level 4 / 5
vars
AudioCue reachedSound
functions / events
OnPlayerEnter() → RespawnPoint.Set(position)

CameraZone
PlayerZone
vars
GameObject zoneCamera (Cinemachine)
functions / events
OnPlayerEnter() → camera on
OnPlayerExit() → camera off
SharedSfx.Play

GameplayClock
MonoBehaviour
vars
TMP_Text label
float Elapsed
functions / events
Update() → "PLAY TIME" counter

TutorialHint
PlayerZone
vars
string message
bool once
functions / events
OnPlayerEnter() → ShowHint
OnPlayerExit() → HideHint

creates
disable / re-enable player
holds
ShowHint() / HideHint()
CompleteGame()
Config / Music / Save()
PlayTransition()
Volume / Music / Sfx
polls Player position
SharedSfx.Play
SetVolume() / SetBrightness()
+= BrightnessChanged

Audio System Smoking.Audio

AudioBootstrap
Composition root · DontDestroyOnLoad
vars
AudioMixer mixer
AudioMixerGroup musicGroup, sfxGroup
int maxSfxVoices = 32
static AudioBootstrap active
functions / events
IMusicPlayer Music
ISfxPlayer Sfx
IAudioVolumeControl Volume
static ISfxPlayer SharedSfx
Awake() → builds the 4 services
Update() → pool / music Tick

SfxPlayer
ISfxPlayer
functions / events
Play(cue)
PlayAt(cue, position)
PlayFollowing(cue, target)
StopAll(fade)

MusicPlayer
IMusicPlayer
vars
Track[2] // crossfade
functions / events
Play(cue, fade)
Stop(fade) / Pause() / Resume()
Tick(dt)

AudioSourcePool
IAudioRentalOwner
vars
List<Voice> voices
int capacity
functions / events
Play(cue, pos, target, following, force2D) → AudioHandle
Tick(dt)

AudioVolumeController
IAudioVolumeControl
vars
AudioMixer mixer
float[] volumes, bool[] muted (PlayerPrefs)
functions / events
GetVolume(bus) / SetVolume(bus, v)
SetMuted(bus, muted)
ApplyAll()
Save()

AudioCue
ScriptableObject
vars
AudioClip[] clips
float volume, Vector2 pitchRange
bool loop
float spatialBlend
int maxInstances
functions / events
TrySample(out clip, gain, pitch)
Configure(source, ...)

AudioHandle
readonly struct
functions / events
IsValid
Stop(fade) / Pause() / Resume()

new
new
new
Play()
plays
TrySample() / Configure()
returns

Environment (decoration) Smoking.Environment

Puzzle · Rotating Stairs Smoking.Puzzle

ClickInput
MonoBehaviour · on Main Camera
vars
Camera viewCamera
LayerMask clickMask
float maxDistance
functions / events
Update() → mouse press → raycast
→ GetComponentInParent<IClickable>()

IClickable
interface
functions / events
OnClicked(RaycastHit hit)

MagicCube
MonoBehaviour · IClickable
vars
RotatableStair target
Transform triggerX, triggerY, triggerZ
Collider cube
float angle = -90
AudioCue rotateSound
functions / events
OnClicked(hit)
FindTrigger(hit) → face → axis

RotatableStair
MonoBehaviour
vars
float duration
Quaternion targetRotation
bool IsRotating
functions / events
TryRotate(worldAxis, angle)
static Turn(target, from, to, duration, done)

OnClicked(hit)
TryRotate(axis, -90)

Puzzle · Phantom Stairs & Books Smoking.Puzzle

BookSwarm
MonoBehaviour · book area
vars
Vector3 areaSize
Transform player
PhantomSegment[] segments
int answerCopies
List<DecoyBooks> decoys
GameObject[] bookPrefabs
List<Book> books
bool active
Vector3 Extents
functions / events
LateUpdate() → FollowCamera, Release / Dismiss
HandleClick(book)
Spawn(color)
bool Solved

Book
MonoBehaviour · IClickable · spawned
vars
BookSwarm swarm
Color Tint
float speed
bool busy
functions / events
Init(owner, color, ...)
OnClicked() → swarm.HandleClick
FlyTo(destination, arrived)
Reject() / Leave()

PhantomSegment
MonoBehaviour · can sit under RotatableStair
vars
Color requiredColor
Material phantomMaterial, solidTemplate
Transform hint
bool IsSolid, IsWaiting
functions / events
Accepts(color)
Claim()
Solidify() → colliders on
SetSolid(solid)

Spawn
HandleClick(this)
Accept
Turn(lever)
Solidify()
child of (optional)

Puzzle · Shadow Puzzle Smoking.Puzzle

ShadowPuzzle
MonoBehaviour · sequence controller
vars
Vector3 areaSize
PathWalker player
ShadowWall wall
Light spotlight
List<ShadowShape> shapes
GameObject shadowCamera
Transform lever
float moveSpeed, sideRange, depthRange
float stepRatio, failDepth, respawnDelay
State state (Waiting / Playing / Done)
ShadowWalker walker
Vector3 lightStart
functions / events
Update() → trigger, MoveLight, Rebuild, Tick
Begin() → dim world, CreatePuppet()
MoveLight(dt) → WASD
HandleFell() → Respawn() (reset light)
HandleGoal() → Finish() lever, lights on
ApplyDarkness(k)
HideClutter() / ShowClutter()
FollowShadow(dt)

ShadowWalker
MonoBehaviour · on shadow puppet
vars
ShadowWall wall
Transform lightSource
IReadOnlyList<ShadowShape> shapes
Vector2 position // wall space
int direction
bool falling, float fallSpeed
functions / events
event Action Fell, ReachedGoal
Init(...) / Spawn()
Tick(dt) → Walk / Fall
Ground() / Inside() / Blocked()
Place() → puppet on light ray

ShadowShape
MonoBehaviour · shadow caster
vars
bool solid, goal
List<Vector2> Outline
functions / events
Rebuild(wall, light) → project + ConvexHull
SpanAt(x, out bottom, out top)

ShadowWall
MonoBehaviour · the 2D world
vars
Vector3 localNormal
Origin / Normal / Up / Right
functions / events
ToWall(world) / ToWorld(point)
TryProject(light, point, out shadow)
Magnification(light, point)

CreatePuppet
Fell / ReachedGoal
Tick()
Rebuild(wall, light)
SpanAt(x)
TryProject() / ToWorld()
TryProject()

UI / Menus Smoking.GameFlow

MenuPanels
MonoBehaviour · Canvas
vars
bool gameplay
GameObject homePanel, settingsPanel, pausePanel, levelSelectPanel
bool IsPaused
functions / events
HandleEscape()
Pause() / Resume() → Time.timeScale
OpenSettings() / CloseSettings()
OpenLevelSelect() / CloseLevelSelect()

UI Buttons
Unity Button.onClick (set in Inspector)
functions / events
Start / Level Select / Restart / Menu / Quit → SceneFlow
Settings / Resume / Back → MenuPanels

SettingsPanel
MonoBehaviour
vars
Slider musicSlider, sfxSlider, brightnessSlider
AudioCue previewSound
functions / events
OnEnable() → read settings
SetMusic(v) / SetSfx(v)
SetBrightness(v)
PreviewSfx()

MenuWalker
MonoBehaviour · main menu decoration
vars
RectTransform route
RawImage view
Camera filmCamera
float height, walkSpeed
functions / events
LateUpdate() → FitCamera(), Walk()
Ground(x)

CreditsScroller
MonoBehaviour · Credits scene
vars
TMP_Text creditsText
float pixelsPerSecond
bool loop, Finished
functions / events
RestartRoll()
Update() → scroll

BrightnessOverlay
MonoBehaviour
vars
Image shade
functions / events
Apply(brightness)

onClick
onClick
SharedSfx.Play
SharedSfx.Play

LightLinkedFade
MonoBehaviour
vars
Light source
functions / events
LateUpdate() → dim glow with light intensity

SpaceshipTraffic
MonoBehaviour
vars
GameObject[] shipPrefabs
int count
Vector2 speedRange
Light dimWith
functions / events
Launch() / Fly() / Land()

Floater
MonoBehaviour
vars
float bobHeight, bobPeriod, spinSpeed
functions / events
Update() → bob + spin

dims world lights (indirect)